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SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

“SUBSTITUTION VS TRANSPOSITION: WHICH PROVIDES BETTER SECURITY

CSA5115-CRYPTOGRAPHY AND NETWORK SECURITY

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INTRODUCTION

SUBSTITUTION AND TRANSPOSITION

- Substitution and Transposition. Every single time we swipe a credit card, log into our banking apps, or send a private text message, digital systems rely on these two distinct concepts to scramble our data and hide it from hackers.
- While one changes the identity of the information and the other shuffles its structure, understanding the critical difference between them reveals exactly how modern cybersecurity keeps our live data safe.

DIFFERENCE SUBSTITUTION AND TRANSPOSITION

SUBSTITUTION AND TRANSPOSITION

1. The Core Mechanism

- Substitution changes the identity of the data while keeping the positions identical.
- Transposition changes the location of the data while keeping the identities identical.

2. Cryptographic Purpose (Real-Time)

- Substitution creates Confusion. It hides the relationship between the encryption key and the ciphertext.
- Transposition creates Diffusion. It spreads the influence of a single plaintext character across the whole ciphertext.

Conclusion

Why Transposition is Safer

- It destroys structure: Transposition shatters word shapes and language patterns by shuffling the positions of the letters.* Substitution is easily cracked: Simple substitution keeps words in their exact original order. Hackers can easily break it in seconds using frequency analysis (counting how often letters like 'E' or 'T' naturally appear).

THANK YOU